Multitask Semantic Boundary Awareness Network for Remote Sensing Image Segmentation

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***Abstract*— In remote sensing images, boundary information plays a crucial role in land-cover segmentation. However, it is a challenging problem that sufficiently extracts complete and sharp boundaries from complex very-high-resolution (VHR) remote sensing images. To tackle this problem, we propose a semantic boundary awareness network (SBANet). The SBANet captures refined boundary information of land covers in feature extraction and then supervises its learning with a designed boundary loss. The key of SBANet includes boundary attention module (BA-module) and adaptive weights of multitask learning (AWML). The BA-module is proposed to capture land-cover boundary information from hierarchical features aggregation in a bottom-up manner. It emphasizes useful boundary information and relieves noise information in low-level features with the guidance of high-level features. To directly learn the boundary information, AWML adds a boundary loss to the original semantic loss by an adaptive fusion manner. This multitask learning enables the semantic information and the boundary information to work collaboratively and promote each other. Note that the BA-module and AWML are plug-and-play. Experimental results demonstrate the effectiveness of the proposed SBANet on the available ISPRS 2-D semantic labeling Potsdam and Vaihingen data sets. The SBANet also achieves the state-of-the-art performance in terms of overall accuracy (OA) and mean F_1 score (m-F_1).**

***Index Terms*— Boundary attention, multilevel aggregation, multitask learning, remote sensing, semantic segmentation.**

I. Introduction

With the rapid development of earth space science and satellite sensor technology, remote sensing image interpretation has opened new horizons [1]. Many research works and tasks, such as object detection [2], image matching [3], and semantic segmentation [4], are closely related to it. Semantic segmentation is a critical task, making remote sensing images better understood and analyzed [5]. Semantic segmentation aims at assigning the corresponding semantic label to each pixel in remote sensing images. The generated pixelwise labels are applied to scene maps reconstruction, intelligent agriculture, and urban planning.

In remote sensing image segmentation, the commonly used methods can be divided into two groups: traditional region merging methods and convolution neural network (CNN) methods. The traditional region merging methods extract handcrafted features of remote sensing images, whereas the CNN methods extract robust features through constructing segmentation networks.

Generally, the traditional region merging methods include two processes: the initial image oversegmentation and the optimal segmentation. First, in the initial oversegmentation, the original remote sensing images are oversegmented to obtain small regions grouped by neighboring pixels. Then, in the optimal segmentation, the most similar adjacent regions are merged until the termination condition is reached. The simple linear iterative cluster (SLIC) algorithm [6] is a common oversegmentation method. In the optimal segmentation, the Markov model [7] and region adjacency graph [8] model are common merging methods. The effectiveness of the region merging method greatly depends on setting a reasonable merge standard, which reflects similarity measures for adjacent regions. Maxwell *et al.* [9] and Ton *et al.* [10] measured similarities by spectral features in remote sensing images with low or medium spatial resolution. However, the spectral features are rather complicated in high-spatial-resolution images. Thus, these methods lead to a larger intraclass variance and smaller interclass variance [11]. It is not effective to only use spectral features for segmentation. Lu *et al.* [12] and Yang *et al.* [13] extracted texture features for segmentation in high-spatial-resolution images. However, these handcrafted features are expensive and require a special design. How to automatically extract discriminative features, especially in complex scenes, is also a challenge.

Recently, CNN-based semantic segmentation methods have made great progress in natural scenes. Reference [14] is a pioneer that constructs a fully convolutional network (FCN) to generate pixelwise labels in an end-to-end manner. Navab *et al.* [15], Zhou *et al.* [16], Badrinarayanan *et al.* [17], and Drozdzal *et al.* [18] built a U-shaped, encoder–decoder network to extract dense semantic features. The encoder network continuously shrinks image spatial resolution to extract dense features by convolution and pooling operations. While the decoder network gradually recovers spatial

Manuscript received May 17, 2020; revised September 14, 2020 and November 28, 2020; accepted January 6, 2021. Date of publication February 2, 2021; date of current version December 3, 2021. This work was supported in part by the State Key Program of National Natural Science of China under Grant 61836009; in part by the National Natural Science Foundation of China under Grant U1701267; in part by the National Natural Science Foundation of China under Grant 61871310; and in part by the Fund for Foreign Scholars in University Research and Teaching Programs (the 111 Project) under Grant B07048. *(Corresponding author: Licheng Jiao.)*

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Digital Object Identifier 10.1109/TGRS.2021.3050885

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resolution to obtain pixelwise predictions through upsampling operation.

Since much progress has been made, the above CNN-based methods are naturally introduced to remote sensing scenes. Different from the natural images taken in a close range, the scale of remote sensing images is inconsistent. For small-scale land covers, their features may be lost with the decrease of spatial resolution, which leads to poor segmentation results. Considering the characteristic of remote sensing images, some research works construct networks based on multiscale feature aggregation. Hamaguchi *et al.* [19] and Nogueira *et al.* [20] employes dilated convolution [21] to enhance context information for feature aggregation. Audebert *et al.* [22] constructed multikernel convolution layers for multiscale predictions. Different from the above expensive multikernel convolutions, Maggiori *et al.* [23], Marmanis *et al.* [24], and Sun *et al.* [25] merged the multiscale features from distinct (shallow or deep) layers. In particular, Marmanis *et al.* [24] and Sun *et al.* [25] designed an additional branch, which uses digital surface model (DSM) data as input to improve image segmentation.

The above research works mainly directly fuse low-level features with high-level features. It is worth noting that low-level features contain fruitful detailed information, whereas high-level features contain more discriminative semantic information. We define the difference between low- and high-level features as semantic gap in this article. Regardless of the semantic gap, direct fusion inevitably embeds background noise of low-level features. This noise influences feature robustness and may cause the detailed information loss. Therefore, it is not quite reasonable to directly fuse these multilevel features. How to relieve the noise information is vital when bridging the semantic gap. In addition, the upsampling operation is often used to generate the pixelwise labels, which may damage boundary information and then lead to blurred segmentation. The boundary information is beneficial to improve precise segmentations. However, its importance is not yet well-considered during the training process.

Based on the above considerations, we propose a semantic boundary awareness network (SBANet) that effectively generates and applies the refined land-cover boundaries. The contributions of the proposed SBANet can be summarized as follows.

1) We propose a boundary attention module (BA-module) that gradually refines complete and sharp boundaries of land covers from the aggregation of multilevel features. The BA-module focuses on relieving noisy information through an adaptive weighted fusion manner for the existing semantic gap in multilevel aggregation.
2) To pertinently learn boundary information, we design a boundary loss to supervise the boundary prediction task. Furthermore, we propose an adaptively weighted strategy to the fusion of boundary loss and the original semantic loss. This multitask learning will jointly learn the semantic information and boundary information to promote the segmentation results and network generalization.

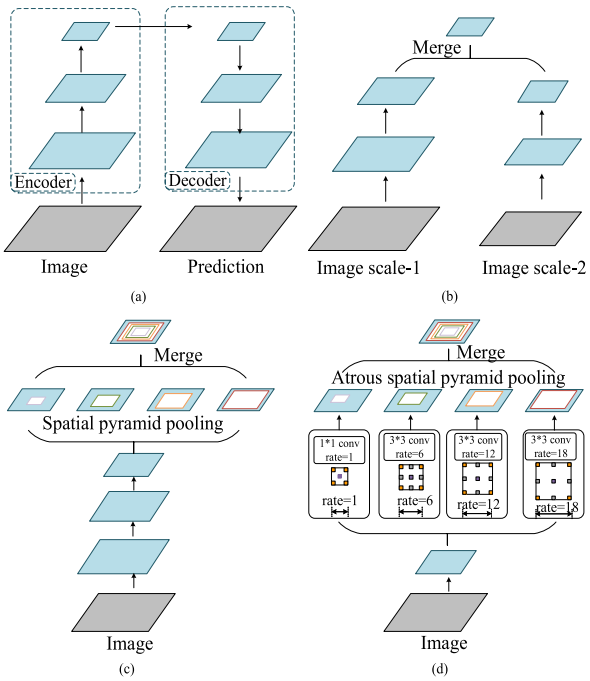


Fig. 1. (a)–(d) Classic U-shaped structure, image pyramid structure, spatial pyramid pooling structure, and ASPP structure, respectively. The rate in (d) refers to the dilated rate in dilated convolutions and denotes the sampling interval on the feature map in convolution.

3) We build an SBANet framework that aims to extract and apply the complete and sharp boundaries of very-high-resolution (VHR) remote sensing images. The SBANet is a multitask learning architecture that emphasizes the role of boundary information in semantic segmentation.

The remainder of this article is organized as follows. Section II shows related works about this article. Section III describes the proposed network. Section IV presents the experiments about the effectiveness of our proposed network and comparisons with other methods. Finally, in Section V, we conclude this article.

II. Related Work

A. Multilevel Features Ensembling

Recent research works focus on multiscale/multilevel features ensembling to improve the performance of semantic segmentation. The U-shaped structure, as shown in Fig. 1(a), is commonly used for segmentation. Zhang *et al.* [26] attempted to insert more semantic information into low-level features and embedded more spatial information into high-level features. Lin *et al.* [27] recursively refined low-resolution semantic features in a top-down manner. Pyramid structure is widely used due to its utilization of multiscales features. Lin *et al.* [28] constructed multiscale CNNs and pyramid pooling to obtain context for patch background. The multiscale CNNs are image pyramid structures that take multiscale images in multibranch as a network input, as shown in Fig. 1(b). Compared to the great consumption of image pyramid, Zhao *et al.* [29] proposed a feature pyramid pooling

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module that incorporates multiscale context by aggregating features at different scales. While the Deeplabv2 [30] and its variants Deeplabv3 [31], Deeplabv3+ [32] propose the atrous spatial pyramid pooling (ASPP) module by combining dilated convolution [21] with spatial pyramid pooling [33] shown in Fig. 1(c). The ASPP, as shown in Fig 1(d), fuses multiscale features from atrous convolutions with different dilated rates in a parallel manner. For clarity, assuming 1-D input signals $x[i]$, the output $y[i]$ of atrous convolutions with $x[i]$ is defined as

$$y[i] = \sum_{k=1}^{K} x[i + r \cdot k] \cdot w[k] \tag{1}$$

where r denotes the expansion rate that determines sample stride of input signals and $w[k]$ is a certain filter, such as 3×3 filter.

Several works [34]–[38] employ graph model [e.g., conditional random fields (CRFs)] to construct semantic information, especially on object boundaries. Dense-CRF [39] uses CRFs on network outputs, instead of incorporating CRFs into a network. This way may introduce increased costs. CRFasRNN [40] indeed incorporates CRFs into the network to generate end-to-end structural predictions. Compared to CRFasRNN, Contextual-CRF [41] constructs the spatial semantics more accurately and gets smoother predictions. However, the above CRF-based models may fail to capture semantic relationships between the pixels, which causes spatially disjoint predictions. Gadde *et al.* [42], Ke *et al.* [43], and Liu *et al.* [44] worked on extracting affinity between the adjacent pixels for structured information propagation. Cheng *et al.* [45] directly concatenated multiresolution features, while LRR [46] builds a multilevel feature reconstruction architecture. LRR successively reconstructs low-resolution features by multiplicative gating and skip connection. The above research works focus on enhancing the context information and obtaining refined boundaries based on multiscale feature aggregation.

B. Boundary Detection

Boundary detection is a fundamental component of many computer vision tasks. Some research works focus on extracting low-level boundaries using boundary detection operators. For example, first-order boundary detection operators, Canny [47] and Heath *et al.* [48], calculate the first-order differential of images to detect sharp changes. However, the saddle point may exist in the first-order differential, and the Laplacian of Gaussian (LoG) [48] operator captures boundaries by calculating the second-order differential. Recently, some researchers have improved the performance of boundary detection by CNNs. HED [49] learns rich hierarchical boundary representations and uses side supervision to guide its learning. CASENet [50] regards boundary detection as a more challenging multilabel problem. Each boundary pixel is associated with more than one label, due to that it appears in contours or junctions belonging to two or more semantic categories. The boundary information is beneficial to generate the refined boundaries in semantic segmentation.

C. Multitask Learning

Multitask learning collaboratively promotes the model generalization and robustness, which shares representations between the primary and auxiliary tasks [51]. Several non-neural algorithms [52]–[54] mainly employ norm regularization to construct a sparse model and relationships between different tasks. Recent CNN-based algorithms use parameter sharings, the hard and soft parameter sharing. The hard parameter sharing is achieved by sharing hidden layers among all tasks while keeping task-specific layers. This parameter sharing method greatly reduces the risk of overfitting. While sharing soft parameters, each task has its own parameters and models. Also, these parameters between different tasks are constrained by normalization. ResUNet-a [55] regards boundary extraction as a multitask segmentation and predicts multiple labels for each pixel. It uses the Tanimoto distance with completions to constrain multitask segmentation. ResUNet-a improves convergence speed without weight initialization. Long *et al.* [56] proposed a task-specific network layer as *a priori* knowledge to learn the relationship between tasks. Compared to the fixed structure in [56], Lu *et al.* [57] dynamically clustered similar models by greedily widening models from a simple structure. However, it ignores the correlation in each task, which may not get the optimal global structure.

D. Weakly Supervised Learning

Weakly supervised learning has been attracting much attention on both natural scene images [58]–[60] and remote sensing images [61]–[63]. Weakly supervised learning directly uses image- or bounding box-level data sets for pixel-wise labeling without high human labor cost data sets. Shi *et al.* [59] formulated the weakly supervised object detection as multiple-instance learning by viewing each training image as a bag and iteratively selecting high-scoring instances from each bag. Cheng *et al.* [58] combined selective search and Gradient-weighted Class Activation Mapping (Grad-CAM) [60] to generate and select high-quality proposals. It is also a challenge for weakly supervised object detection in the large-scale image with complex backgrounds. Yao *et al.* [61] applied a dynamic curriculum learning strategy to progressively learn the object detectors in remote sensing images. Feng *et al.* proposed a contextual instance refinement to discover the whole object rather than the most representative part. Zhou *et al.* trained a robust detector that employs a negative bootstrapping scheme by selecting the most informative negative samples.

III. Proposed Method

A. Boundary Attention Module

Bridging semantic gap is a challenge of semantic segmentation [64]. Semantic gap usually exists in shallow and deep layers. Shallow layers capture more detailed information, such as texture and boundary, whereas deep layers extract more discriminative information, such as semantic features. Based on the above description, multilevel feature aggregation is necessary to make them complementary. However, for multilevel

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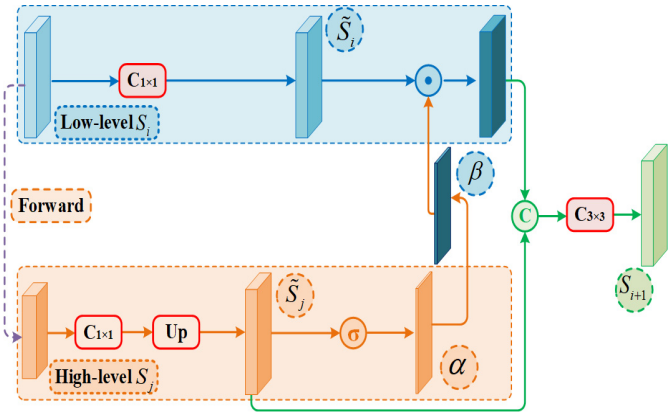


Fig. 2. Structure of the BA-module. The symbols σ and $\odot$, respectively, denote sigmoid function and elementwise product, respectively, and the symbol © represents concatenation. Section III-A describes more details.

features with different importance, it is not quite reasonable to directly fuse features with the same weight. Moreover, direct fusion inevitably brings background noise from low-level features, which may damage feature robustness. In this section, we propose a more efficient module, named BA-module. The BA-module aggregates multilevel features with adaptive weights by attention mechanism. The BA-module enables to relieve the noise information and encourages the segmentation network to focus on complete land-cover feature extraction. Therefore, the BA-module can extract complete and sharp boundaries around land covers. We illustrate the BA-module in Fig. 2.

Given features S_i, S_j (i<j) from adjacent low level and high level, respectively, the BA-module is described as follows.

First, to obtain the higher spatial-resolution features $\tilde{S}_j \in R^{H_1\times W_1\times C_1}$. High-level features $S_j \in R^{H_2\times W_2\times C_2}$ are fed into a convolution and then upsampled to generate higher spatial resolution features $\tilde{S}_j$. H, W, and C represent the height, width, and channel of features, respectively. As well known, deep layers extract more discriminative features with lower spatial resolution. However, extracting boundary information requires pixelwise features with higher spatial resolution. Therefore, we need to expand the spatial resolution of S_j.

Second, to extract the attention mask α of high-level features, deep layers have a larger receptive field than shallow layers. Therefore, they can extract more credible high-level features. Instead, shallow layers capture low-level features with lower credibility. These high-level features can be used as guidance to activate useful information and suppress noise information in low-level features. This way also helps bridge the semantic gap. Therefore, we generate the attention mask of upsampled high-level features $\tilde{S}_j$ by a sigmoid function. The attention mask α is represented as

$$\alpha = \text{Sigmoid}\left(\tilde{S}_j\right) = \frac{1}{1+\exp\left(-\tilde{S}_j\right)} \tag{2}$$

where $\alpha \in R^{H_1\times W_1\times 1}$ values range from 0 to 1. It describes the significance of $\tilde{S}_j$. For α, the larger the value, the more significant the region is, and vice versa.

Third, to generate the attention mask β of lower salient regions, the BA-module aims to extract low saliency regions by erasing the current predicted salient areas in the high-level features $\tilde{S}_j$. The inner regions of land covers contain more abundant texture information, whereas the outer regions contain more abundant boundary information. The same convolution gives different responses to these regions. Based on the network working mechanism, the regions with high responses have high saliency, whereas the regions with low responses have low saliency. These high saliency regions have passive effects on the extraction of low saliency regions. Therefore, we use image residuals (the subtraction of features) to activate low saliency regions by suppressing high saliency regions. With α, we generate the attention mask β of low salient regions as

$$\beta = 1 - \alpha. \tag{3}$$

Fourth, to obtain the attentive output S_{i+1}, $\tilde{S}_i$ is generated by a 1×1 convolution applied to low-level features S_i. $\tilde{S}_i$ keeps the same channel number with high-level features $\tilde{S}_j$. Given the attention mask β, the output features S_{i+1} are calculated as

$$S_{i+1} = \left(\left(\tilde{S}_i \odot \beta\right) \copyright \tilde{S}_j\right) * \omega_t \tag{4}$$

where $\odot$ represents the elementwise product and © represents the concatenation. For S_{i+1}, we first obtain more credible features in $\tilde{S}_i$. It is weighted by attention mask β of high-level features $\tilde{S}_j$. This encourages our network to focus on low salient regions. Then, the weighted features are fused with high-level features $\tilde{S}_j$ in a residual way. It means that the low saliency region and the high saliency region are fused. The fusion enables the network to extract the complete land-cover region and then generate complete boundaries. Moreover, the residual way makes gradients flow smoothly and alleviates the gradient vanishing problem. Finally, it follows a convolution with kernel ω_t for better aggregation. Each convolution is followed by batch normalization and rectified linear unit (ReLU) function in turn.

The BA-module aggregates the multilevel features to bridge the semantic gap using the attention mechanism. With the BA-module, our network extracts more complete land-cover boundary information, as shown in Fig. 5.

B. Adaptive Weights of Multitask Learning

Boundary prediction is beneficial to improve semantic segmentation task [24]. The boundary prediction is a two-category task, while the semantic segmentation is a multicategory task in most cases. Although the existing methods consider the boundary information, they learn the boundary information through semantic loss, without special boundary loss. Considering the above description, we separate boundary information from semantic features and then supervise them with the corresponding losses, L_{bdr} and L_{seg}. Furthermore, the above strategy forms multitask learning, named adaptive weights of multitask learning (AWML).

For semantic features, we apply standard cross-entropy loss function on its predictions. The segmentation loss L_{seg} is

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denoted as

$$L_{\text{seg}} = \sum_{i,j} CE(\hat{y}, y) = -\frac{1}{m}\sum_{j}^{m}\sum_{i}^{C}\left[y_{ji}\log\left(\hat{y}_{ji}\right)+\left(1-y_{ji}\right)\log\left(1-\hat{y}_{ji}\right)\right] \quad (5)$$

where m and C denote the number of samples and categories, respectively. Also, the symbols y and $\hat{y}$ represent the segmentation ground truth (GT) and the segmentation predictions, respectively.

For boundary features, the network generates boundary predictions through the proposed BA-module. The predicted values range from 0 to 1. Here, 0 shows that the pixel is boundary pixels (negative), and 1 shows that the pixel is non-boundary pixels (positive). It is worth noting that the negative samples are much more than the positive samples in Potsdam and Vaihigen data sets. Considering the imbalance data, we weight the loss function by the proportion of positive and negative samples. The boundary GT is generated from the segmentation GT using the classical image processing algorithm, Euclidean distance transform algorithm [65]. Given the predicted boundary map $\hat{b}$ and the boundary GT b for each image, the boundary loss L_{bdr} is described as

$$L_{\text{bdr}} = \sum_{j} BCE\left(\hat{b}, b\right) = -\frac{1}{m}\sum_{j}^{m}\left[\lambda_1 \cdot b_j \log\left(\hat{b}_j\right) + \lambda_2 \cdot \left(1-b_j\right)\log\left(1-\hat{b}_j\right)\right] \quad (6)$$

where

$$\lambda_1 = \psi \cdot \frac{B^0}{B^1+B^0}, \qquad \lambda_2 = \frac{B^1}{B^1+B^0} \quad (7)$$

with B^0 and B^1 the number of nonboundary and boundary pixels, respectively. The parameter $\psi(\psi > 0)$ is used to balance the proportion of B^0 and B^1 in total samples. With the weights λ_1 and λ_2, the loss L_{bdr} gives a higher penalty for negative samples than positive samples. This encourages network to learn boundaries.

Therefore, the multitask learning loss L_{total} is combined by L_{seg} and L_{bdr} as

$$L_{\text{total}} = \eta_1 \cdot L_{\text{seg}} + \eta_2 \cdot L_{\text{bdr}} \quad (8)$$

where

$$\eta_1 + \eta_2 = K \quad (9)$$

with η_1 and η_2 weights of L_{bdr} and L_{seg}, respectively, and $K(K > 0)$ is a constant that controls the scale of these losses.

Manually weighting multiple tasks is a difficult and expensive process. Moreover, if loss weights are fixed during the training phase, a large value loss will be assigned a small initial weight. Later in the training phase, the loss value decreases, and the small weight will limit the loss. As a result, the corresponding task cannot be well learned. Therefore, weights of multiple tasks are required to change continuously, along with the training process. Considering the real time, we propose an adaptively weighting strategy for L_{bdr} and L_{seg} by reviewing the change rate of each loss value over the training process. Assume that t represents the index of each iteration. Therefore, the weights η_1 and η_2 are described as

$$\begin{cases} \eta_1(t) := \dfrac{K \cdot \exp\left(\bigtriangledown_{\text{seg}}(t-1)/\mathcal{T}\right)}{\exp\left(\bigtriangledown_{\text{seg}}(t-1)/\mathcal{T}\right)+\exp(\bigtriangledown_{\text{bdr}}(t-1)/\mathcal{T})} & \text{if } t > 2 \\ \eta_1(t) = 0.5 & \text{if } t = 1, 2 \end{cases}$$
$$\begin{cases} \eta_2(t) := \dfrac{K \cdot \exp\left(\bigtriangledown_{\text{seg}}(t-1)/\mathcal{T}\right)}{\exp\left(\bigtriangledown_{\text{seg}}(t-1)/\mathcal{T}\right)+\exp(\bigtriangledown_{\text{bdr}}(t-1)/\mathcal{T})} & \text{if } t > 2 \\ \eta_2(t) = 0.5 & \text{if } t = 1, 2 \end{cases} \quad (10)$$

where $\exp(\cdot)$ denotes the softmax operator with a hyperparameter $\mathcal{T}(\mathcal{T} > 0)$. $\bigtriangledown_{\text{seg}}(\cdot)$ and $\bigtriangledown_{\text{bdr}}(\cdot)$ are the change rates of L_{seg} and L_{bdr}, respectively, which are calculated as

$$\begin{cases} \bigtriangledown_{\text{seg}}(t-1) = \dfrac{\overline{L_{\text{seg}}}(t-1)}{\overline{L_{\text{seg}}}(t-2)} & \text{if } t > 2 \\ \bigtriangledown_{\text{seg}}(t) = 1 & \text{if } t = 1, 2 \end{cases}$$
$$\begin{cases} \bigtriangledown_{\text{bdr}}(t-1) = \dfrac{\overline{L_{\text{bdr}}}(t-1)}{\overline{L_{\text{bdr}}}(t-2)} & \text{if } t > 2 \\ \bigtriangledown_{\text{bdr}}(t) = 1 & \text{if } t = 1, 2. \end{cases} \quad (11)$$

For η_1 and η_2, we first calculate the change rates of corresponding loss values $\overline{L_{\text{seg}}}$ and $\overline{L_{\text{bdr}}}$ after one iteration as (11). The loss values $\overline{L_{\text{seg}}}$ and $\overline{L_{\text{bdr}}}$ denote the average of their corresponding loss L_{seg} and L_{bdr}, respectively, over several iterations. The average values reduce the uncertainty of gradient descent and maintain the stability of training. We then perform a softmax operation with a fixed value $\mathcal{T}$ to keep gradients flowing smoothly in backpropagation. $\mathcal{T}$ is similar to the Gumbel Softmax [66] that controls the softness of the softmax function. With the parameter $\mathcal{T}$, the output distribution of softmax will be smoother and more uniform, while its relative numerical size will not change. Finally, we multiply the results of softmax operation by K to obtain the weights η_1 and η_2 as (10). K is the same as K in (9). K keeps that their values will not be too small.

The AWML encourages our network to learn the boundary information. In addition, the adaptive fusion strategy dynamically adapts the losses with different scales, which is helpful for jointly learning the segmentation prediction and boundary prediction.

C. SBANet for Segmentation

We construct a multitask learning architecture, named SBANet. Consider the effectiveness of Deeplabv3+ [32], and we take it as our baseline network for the semantic segmentation task. Fig. 3 shows the architecture of the proposed SBANet. The SBANet is mainly composed of encoder, ASPP, and decoder stages.

In the encoder stage, the SBANet takes image $I \in R^{H\times W\times C}$ with height H and width W as input. The image I is fed

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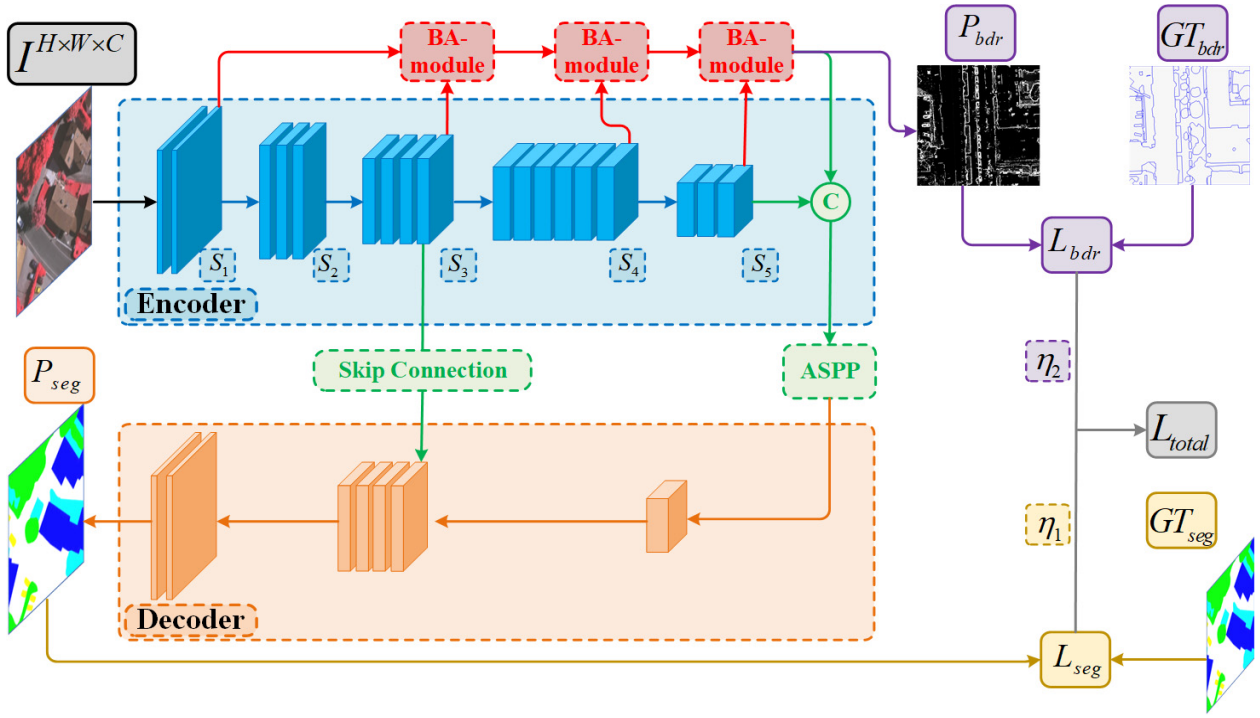


Fig. 3. Structure of the proposed SBANet. The network applies the BA-modules to generate boundary predictions. We jointly supervise boundary predictions and segmentation tasks with a total loss L_{total}, which is combined by segmentation loss L_{seg} and boundary loss L_{bdr} with adaptive weights η_1 and η, respectively. GT_{seg} and GT_{bdr} denote the segmentation GT and boundary GT, respectively. P_{seg} and P_{bdr} denote the segmentation predictions and boundary predictions, respectively. The BA-module is shown in Fig. 2 and detailed description in Section III-B. The ASPP refers to the atrous spatial pyramid pooling mentioned in [30], shown in Fig. 1(d).

into the encoder module to extract dense semantic features through the backbone network. We follow the prior work of Deeplabv3+, which keeps five levels of convolutional blocks to extract dense semantic features in the backbone. Specifically, the first level contains convolution and max pooling. The remaining levels contain three, four, six, and three residual blocks. Each residual block consists of three convolutions with 3×3 kernel size. The output features S_{out} of the final level are represented as

$$S_{\text{out}} = f_{en}(I) \tag{12}$$

where $f_{en}(\cdot)$ notes the convolution operations in the encoder stage.

The BA-module processes the side-out features S_i of these levels in a bottom-up manner. After several BA-modules, we get the final boundary map $\hat{b}$. The map is used to calculate the boundary loss with the binary boundary GT. The concatenation features of the final boundary map $\hat{b}$ and the final dense semantic features S_{out} are fed into the ASPP module [30], as shown in Fig. 1 (d). The ASPP module enlarges multiscale context features with parallel dilated convolutions of different dilate rates. The output of the ASPP module is described as

$$C_0 = \text{ASPP}\left(\hat{b} \copyright S_{\text{out}}\right) \tag{13}$$

where © denotes the concatenation operator.

In the decoder stage, we upsample the low spatial-resolution features C_0 from the ASPP module through bilinear interpolations and convolutions to generate higher spatial-resolution features C_1. The skip connection is employed to reuse the corresponding latent features in encoder stage. This helps our network to recover more high spatial-resolution features and further increases network stability. Given the number of categories C', the output prediction $\hat{y} \in R^{H\times W\times C'}$ is denoted as

$$\hat{y} = f_{de}(C_0) \tag{14}$$

where $f_{de}(\cdot)$ is the convolutions in the decoder stage.

The SBANet promotes bridging the semantic gap lied in multilevel features and generates the refined land-cover segmentations, especially in boundaries.

IV. Experimental Results

A. Experimental Setting

1) Data set: The Potsdam and Vaihingen semantic labeling data sets are released by ISPRS Commission WG II/4. Each data set consists of VHR treue ortho photo (TOP) tiles and the corresponding DSMs derived from dense image matching techniques. Each data set contains six most common land-cover categories, namely impervious surfaces (e.g., roads), building, low vegetation, tree, car, and clutter/background.

The Potsdam semantic labeling data set consists of 38 tile aerial images (6000 × 6000 pixels) with red–green–blue (RGB) channels. Each image has a corresponding DSM with the same spatial resolution of 5 cm. Among them, 24 tiles are available for training and validation and the remaining 14 for testing. We employ cross validation with a threefold

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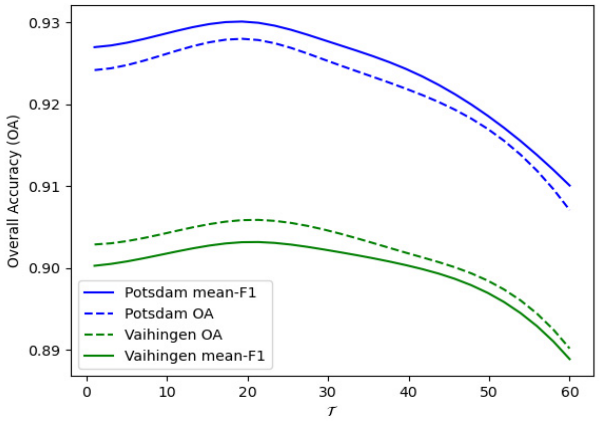


Fig. 4. Curve of effectiveness on AWML at different $\mathcal{T}$ thresholds.

TABLE I
EFFECTIVENESS OF THE BA-MODULE AT DIFFERENT SETTINGS

Dataset	Metric	baseline (%)	number of BA-module			
			1	2	3	4
Potsdam	OA	90.71(0.07)	+0.08(0.02)	+0.46(0.03)	**+0.6(0.04)**	+0.59(0.02)
	m-F_1	89.74(0.05)	+0.07(0.01)	+0.95(0.02)	**+2.0(0.03)**	+1.99(0.01)
Vaihingen	OA	88.60(0.06)	+0.02(0.01)	+0.33(0.01)	**+0.75(0.06)**	+0.74(0.04)
	m-F_1	88.91(0.08)	+0.11(0.02)	+0.38(0.05)	**+0.52(0.03)**	+0.52(0.01)

TABLE II
PERFORMANCE ON DIFFERENT BACKBONE NETWORKS OF OUR NETWORK

Backbone	Potsdam		Vaihingen	
	OA(%)	m-F_1(%)	OA(%)	m-F_1(%)
Resnet-50	91.75(0.04)	91.17(0.10)	90.01(0.08)	89.47(0.09)
Resnet-101	92.80(0.07)	93.01(0.09)	90.59(0.06)	90.34(0.11)

train/validation split that involves 16 tiles for training data and eight tiles for validation.

The Vaihingen semantic labeling data set consists of 33 tile aerial images (approximately 2100 × 2100 pixels) with infrared-red-green (IR-RG) channels. Each image has a corresponding DSM with the same spatial resolution of 9 cm. Among them, 16 tiles with GT labels are used for training and validation, and the remaining 17 images are for testing. We keep threefold train/validation split on cross validation that involves 12 tiles for training data and four tiles for validation.

2) Experimental Settings: We take Deeplabv3+ [32] as our baseline network. Also, ResNet is employed as our backbone network of the SBANet. Both are implemented in PyTorch. We set $\psi = 1$ in our network. Data augmentation with random-scaling and random-flipping operations is adopted on each iteration in the training phase. The input images are randomly cropped into patches of 512 × 512 resolution during each iteration. For Potsdam and Vaihingen, we use a learning rate of $1e^{-}2$ with a polynomial decay policy and a linear warm-up based on the Adam optimizer. In all experiments, the networks run on an NVIDIA GeForce RTX 2080Ti GPU. Each experiment is conducted three times.

3) Evaluation Metrics: We employ overall accuracy (OA) and F_1 score as evaluators for our SBANet. The accuracy evaluator reports the percent of pixels in the image that was correctly classified. The OA is commonly reported for each category separately as well as globally across all categories. The F_1 score evaluator of per category is described as

$$F_1 = \frac{2 * (\text{precision} * \text{recall})}{(\text{precision} + \text{recall})} \tag{15}$$

where precision reports the number of positive category predictions that belong to the positive category. Also, recall reports the number of positive category predictions made out of all positive samples in data sets. The mean F_1 score (m-F_1) denotes the average of F_1 score over all categories.

B. Hyperparameter Setting

1) Hyperparameter $\mathcal{T}$ Setting: In (10), the hyperparameter $\mathcal{T}$ is not robust to segmentation performance, which controls the softness of each weight. Fig. 4 shows the effectiveness of the AWML strategy at different hyperparameter $\mathcal{T}$ thresholds with three BA-modules. It is observed that OA and m-F_1 have a great improvement with $\mathcal{T} = 20$, compared to $\mathcal{T} = 1$. Then, the segmentation accuracy decreases as $\mathcal{T}$ increases. It can be seen from (10), if $\mathcal{T}$ is large enough, the multitask losses weights η_1 and η_2 are close to 1. It means that the two different tasks are weighted equally. However, the two tasks have different scale losses. Also, the task with larger scale loss plays a dominant role in backpropagation. It will induce the network to put too much attention to this task while ignoring another task. Therefore, we choose the optimal $\mathcal{T} = 20$ in all experiments.

2) Input Setting of BA-Module: In our proposed SBANet, the BA-module is used to aggregate low-level features and high-level features to refine land-cover boundaries. To determine which low-level features and high-level features are fused, we experiment on the Potsdam and Vaihingen data sets and record results in Table I. It is worth noting that these experiments are conducted three times, and Table I reports the standard deviations. The encoder module contains five levels of convolutional blocks. For a BA-module, it is used to fuse the first-level features and the fifth-level features. For multiple BA-modules, they are added from high to low levels in a cascaded manner. As shown in Table I, the baseline of one BA-module is slightly better than multiple BA-modules. Large upsampling stride leads to part spatial resolution lost, which has a negative affection on generating boundaries. It also found that adding four BA-modules has no more improvement. For four BA-modules, the fourth BA-module is added between the first level and the second level. Features of the two levels have not apparent changes. To reduce model complexity, we employ three BA-modules in SBANet.

3) Backbone Setting: Table II records the segmentation accuracy of SBANet in different backbones, Resnet-50 and Resnet-101. It is noticed that Resnet-101 achieves better results than Resnet-50. Due to the deeper network of Resnet-101, our network extracts more semantic information. Therefore, we employ Resnet-101 as our backbone network.

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TABLE III
NUMERICAL COMPARISONS WITH STATE OF THE ARTS ON THE POTSDAM TEST SET

Method	Imp. surf.	Building	Low veg.	Tree	Car	m-F_1	OA
FCN [14]	87.85	94.91	81.53	83.77	85.60	86.73	87.06
U-net [15]	88.61	93.29	83.29	85.83	88.47	87.91	86.59
SegNet [17]	88.96	93.72	86.25	84.91	89.17	88.60	87.93
PSPNet [29]	91.61	96.30	86.41	86.84	91.38	90.53	89.45
RefineNet [27]	88.12	94.28	88.73	86.54	89.94	89.52	87.24
CCEM [24]	89.74	94.52	86.88	85.64	91.95	89.75	89.87
SegNet-RC [67]	91.31	95.92	86.24	85.23	91.80	90.10	89.04
CNN+RF+CRF [68]	91.84	96.37	86.97	86.46	92.70	90.39	89.29
Deeplabv3+ [32]	92.14	96.76	85.01	86.08	93.57	90.71	89.74
SBANet	**93.83±0.11**	**98.06±0.10**	**88.97±0.10**	**89.48±0.12**	**94.71±0.09**	**93.01±0.10**	**92.80±0.12**

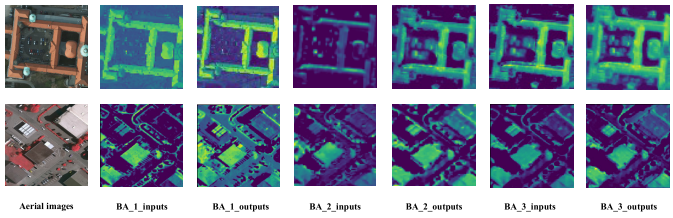
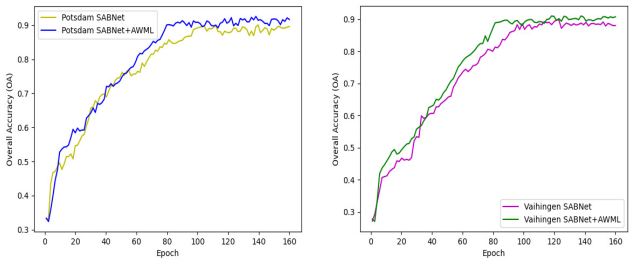


Fig. 5. Visualizations of input and output features from three BA-modules, respectively. For one BA-module, the input image represents high-level features from the backbone network, and the output image represents the output features from the BA-module. (From Top to Bottom) First and second rows demonstrate the Potsdam and Vaihingen train set, respectively.



(a) (b)

Fig. 6. Curves show the effectiveness of AWML. The smoothed curves record the OA value during the training phase. (a) shows the OA values on the Potsdam dataset, while (b) shows the OA values on the Vaihingen dataset.

C. Module Effectiveness Evaluation

1) BA-Module Evaluation: To verify the effectiveness of the BA-module, we provide some visualization examples in Fig. 5. It shows some feature visualizations from inputs and outputs of all BA-modules. We can notice that how BA-modules fuse low- and high-level features. For the second BA-module, the partial regions of the building are not highlighted, which means that these regions are less salience in inputs. The BA-module pays attention to the less salient regions, which complement the salient regions to generate the whole land-cover regions. As shown in outputs, the whole building is highlighted. Based on the above analysis, the BA-module is effective.

2) AWML Evaluation: We also verify the effectiveness of the proposed AWML strategy. Fig. 6 shows the effectiveness curves of how the AWML strategy works during the training phase. We can notice that the *SBANet+AWML* curves reach a steady state faster than the *SBANet* curves. It means that the AWML strategy accelerates the network convergence. It can also be seen that the AWML strategy brings improvements in terms of OA on both training sets. As described in Section III, the AWML strategy enables the SBANet to jointly the learn segmentation task and boundary task. For segmentation task, boundary task is regarded as noise in its training, which contributes to enhancing the robustness of segmentation task. Based on the above analysis, the AWML strategy is effective.

D. Comparison With State-of-the-Art Methods

1) Effectiveness on Potsdam Data Set: We compare the segmentation performance of SBANet with some state-of-the-art methods on the Potsdam test set. Note that the covered methods use the same backbone network (Resnet-101) with three-channel RGB images as inputs. Moreover, we follow their own optimal hyperparameter settings in the comparison methods. The results are shown in Table III. It is observed that our network outperforms other remote sensing image segmentation methods in OA and mean F_1 score (m-F_1).

For FCN, the large-scale Building is well predicted, whereas the small Tree and Car are not well predicted. This is not only because the Car and Tree categories have fewer pixels than other categories but also FCN has no much-paid attention to detailed information. Segmentations by U-net and SegNet are improved with the reuse of detailed features. U-net fuses the shallow-layer features and deep-layer features to refine the details. SegNet upsamples features by calling the max-pooling indices, which records the location of maximum in pooling. The results of RefineNet are better than U-net and SegNet due to its chained residual pooling module. The module captures

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TABLE IV
NUMERICAL COMPARISONS WITH STATE OF THE ARTS ON THE VAIHINGEN TEST SET

Method	Imp. surf.	Building	Low veg.	Tree	Car	m-F_1	OA
FCN [14]	88.67	90.83	76.32	86.67	74.21	83.34	86.51
Unet [15]	89.54	91.06	78.81	89.08	78.73	85.44	87.03
SegNet [17]	86.97	91.21	80.24	84.36	89.17	86.51	85.87
PSPNet [29]	94.32	91.42	81.53	87.91	88.02	88.65	88.99
RefineNet [27]	88.80	91.84	**83.57**	86.62	87.04	87.57	86.93
CCEM [24]	91.03	90.22	82.01	86.89	87.78	87.59	86.32
SegNet-RC [67]	91.40	90.57	81.96	88.43	87.51	87.97	88.80
CNN+RF+CRF [68]	92.91	91.13	82.14	88.77	88.20	88.63	88.72
Deeplabv3+ [32]	94.34	91.35	81.32	87.84	88.14	88.60	88.91
SBANet	**94.36±0.09**	**92.91±0.11**	83.44±0.08	**89.58±0.08**	**91.43±0.10**	**90.34±0.09**	**90.59±0.12**

the background context to refine blurred segmentation, especially on high-spatial-resolution images. In addition, PSPNet and Deeplabv3+ are superior to RefineNet in the large-scale Impervious surface and Building categories. This benefits from the enlarged receptive fields, which are derived from their well-designed multiscale information fusion structure. Both CCEM [24] and SegNet-RC [67] directly fuse the multilevel features while ignoring their semantic gap. Direct fusion may introduce noise information, which has a negative effect on the segmentation results. In addition, CCEM extracts boundary information from the height maps and treats the boundary as an additional channel for feature fusion. Therefore, for small-scale land covers such as Tree and Car, CCEM has better predictions than SegNet-RC. CNN+RF+CRF [68] combines the outputs of random forest and CNNs to refine predictions and improves the accuracy evenly for all categories. It indicates that the handcrafted features indeed contain information that is independent of CNN features.

For the SBANet, the F_1 score of Building, Tree, and Low vegetation has more obvious improvements than Deeplabv3+ (baseline). The training sampler usually crops inputs to obtain small patches of high-spatial-resolution remote sensing images. Thus, the Building with large scale (high spatial resolution) is easily cropped into several regions, which may cause misclassifications in these separated regions. However, our network generates binary boundary maps, which focus on whether the pixels belong to boundaries rather than a certain category. From the boundary view, the separated regions are regarded as a coherent region, which helps recover the error labels. For the Tree and Low vegetation, they have high interclass similarity, such as color and texture. Furthermore, they are usually located in adjacent regions. Therefore, it is a challenge to correctly classify these two categories, especially in the border of adjacent regions. The SBANet guides pixels in the adjacent regions to learning sharper boundaries of the Tree. Hence, the Tree is distinguished from the Low vegetation.

2) Effectiveness on Vaihingen Data Set: We also compare the segmentation performance of SBANet with these state-of-the-art methods on the Vaihingen test set. Table IV records the results. Note that we take the three-channel (IR-RG) images as inputs. The SBANet also outperforms other comparison methods in OA and m-F_1. The Tree and Car categories have a great improvement. However, the segmentation of Low vegetation is worse than that of RefineNet since image channels of Vaihingen data set. For the input images with IR-RG channels, Low vegetation and Tree categories display almost the same red color, which may cause the network to distinguish these two categories not well. In particular, the results of most comparison methods on the Vaihingen data set are inferior to those of Potsdam data set, due to the small number of training samples in the Vaihingen data set.

Figs. 7 and 8 show the visual segmentations of some images obtained by the SBANet and Deeplabv3+ (baseline), respectively. For Deeplabv3+, the Tree category is predicted with blurred boundaries, while predictions of the SBANet have sharper boundaries of the Tree. Moreover, the SABNet reduces some false positive predictions of the Tree, as shown in Fig. 8. For our network, the results of the Building category achieve more coherent labels of precise boundaries. In addition, Fig. 9 shows the visualizations of boundary predictions on the Potsdam and Vaihingen test sets. In boundary labels, blue lines denote land-cover boundaries, whereas in boundary predictions, white line represents the land-cover boundaries. It shows that our predictions have complete and sharp boundaries, especially in the Building and Car categories.

3) Ablation Study: To ensure the contribution of the BA-module and the AWML strategy for the final results, we further conduct experiments, as shown in Table V. It records how our network successively improves the segmentation performance. For SBANet-1, it achieves better results than Deeplabv3+ since the BA-module tends to extract complete land-cover segmentation with precise boundaries. We further use two specific losses L_{seg} and L_{bdr} with the same weights in SBANet-2. It obtains the significant improvements comparing with SBANet-1. Moreover, the results of SBANet-3 are superior to those of SBANet-2, due to the adaptive weights η_1 and η_2. We get the final results as the SBANet-3 on the Potsdam and Vaihingen test sets.

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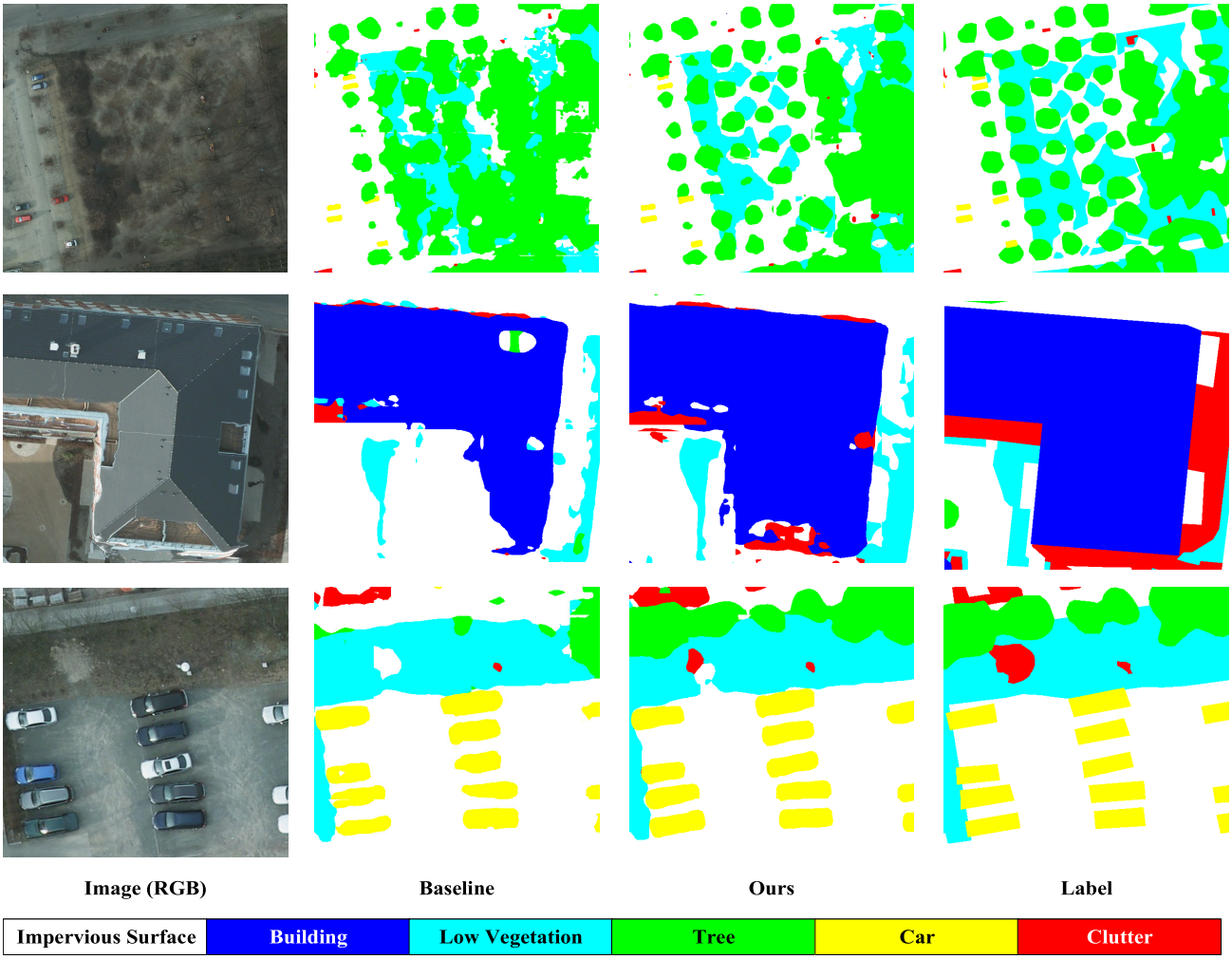


Fig. 7. Qualitative comparisons between our method and baseline on the Potsdam test set. Considering that the image resolution is too large, images are cropped into patch of 1500 × 1500 resolution. (From Top to Bottom) Rows show the crop image "top_potsdam_5_13_RGB," "top_potsdam_3_14_RGB," and "top_potsdam_6_15_RGB." (From Left to Right) Columns demonstrate the aerial images with RGB channels, the predictions of baseline, the predictions of our network, and the image labels.

TABLE V

EFFECTIVENESS OF THE BA-MODULE AND THE AWML STRATEGY ON SBANET IN THE POTSDAM AND VAIHINGEN TEST SET. ADAP. IS THE ABBREVIATION OF ADAPTIVE

Method	BA-module	AWML (Equal Weights)	AWML (Adapt. Weights)	Potsdam		Vaihingen	
				m-F_1(%)	OA(%)	m-F_1(%)	OA(%)
Baseline				90.71	89.74	88.60	88.91
SBANet-1	√			91.74	91.31	89.43	89.35
SBANet-2	√	√		92.53	92.28	89.86	90.17
SBANet-3	√	√	√	**93.01**	**92.80**	**90.34**	**90.59**

4) Complexity Analysis: We compare the complexity of our method with other state-of-the-art methods, as shown in Table VI. For fairness, we record the average inference time of 100 image patches (512 × 512 pixels). Compared with FCN, U-net and SegNet reduce the model capacity by feature reuse. The reuse also reduces the time complexity. PSPNet adopts a pyramid pooling module to enlarge receptive fields while keeping a small increase in model complexity than U-net and SegNet. Different from U-net, RefineNet uses chained residual pooling for upsampling, which makes feature reuse more effective. The pooling module brings a lot of growth in model and time complexity. The space and time complexity of Deeplabv3+ are smaller than those of RefineNet. It may be due to the ASPP. The ASPP enlarges the receptive fields

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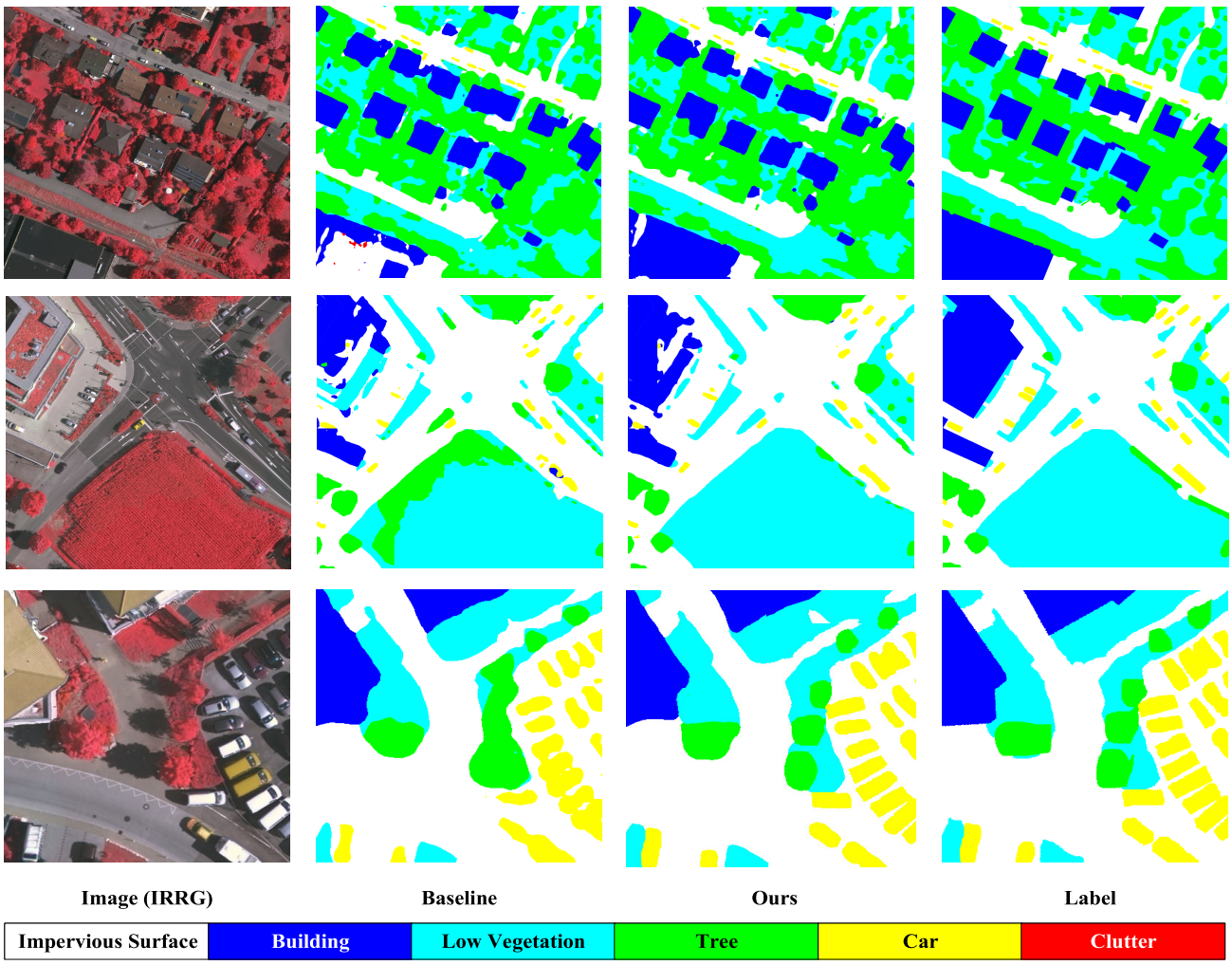


Fig. 8. Qualitative comparisons between our method and baseline on the Vaihingen test set. Considering that the image resolution is too large, images are cropped into patch of 1000 × 1000 resolution. (From Top to Bottom) Rows show the crop images "top_mosaic_09cm_area12," "top_mosaic_09cm_area33," and "top_mosaic_09cm_area2." (From Left to Right) Columns demonstrate the aerial images with IRRG channels, the predictions of baseline, the predictions of our network, and the image labels.

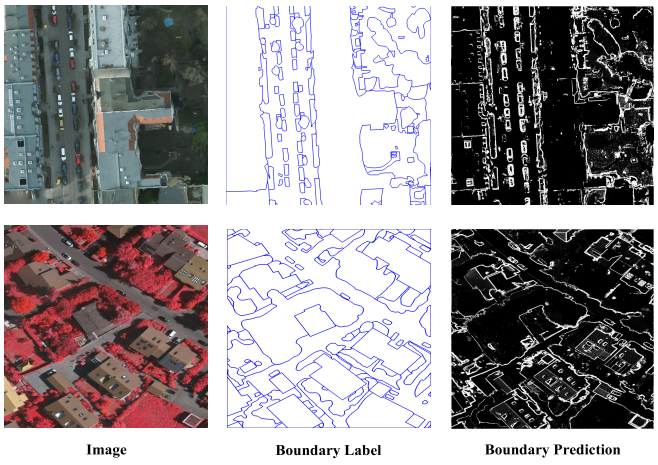


Fig. 9. Visualization of boundary labels and boundary predictions on the Potsdam and Vaihingen test sets. The images are cropped into patches of 1500 × 1500 resolution for seeing clearly. (From Top to Bottom) Rows show the cropped image "top_potsdam_7_3_RGB" and "top_mosaic_09 cm_area15."

without more parameters addition. The SBANet costs more space complexity, while it takes about only 2 s more than Deeplabv3+. The extra inference time of the SBANet is mainly due to the calculation of high-level attention maps. It brings considerable accuracy improvements. Considering the tradeoff between accuracy and time complexity, we employ the SBANet for segmentation.

TABLE VI
COMPARISON OF SPACE COMPLEXITY AND TIME WITH STATE-OF-THE-ART MODELS

Models	Model Size (M)	Time (s)
FCN [14]	537	26
U-net [15]	213	16
SegNet [17]	209	16
RefineNet [27]	454	24
PSPNet [29]	262	18
Deeplabv3+ [32]	437	23
SBANet	524	25

V. CONCLUSION

In this article, we propose the SBANet that makes full use of the aggregation of multilevel features to generate complete

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and sharp land-cover boundaries. Experiments show that the BA-module is effective in extracting the complete boundary information. The AWML is also verified on the challenging ISPRS 2-D semantic labeling Potsdam and Vaihingen data sets. However, the SBANet is sensitive to hyperparameters, which affects network stability and feature robustness. In addition, the SBANet contains more network parameters, which may include redundant parameters. In the near future, we will explore the robust hyperparameters in an adaptive way. Moreover, we will also prune the network to reduce the network parameters while ensuring network performance.

ACKNOWLEDGMENT

The authors would like to thank the providers of the available data sets. The ISPRS Potsdam Challenge data set and the Vaihingen Challenge data set are produced by the International Society for Photogrammetry and Remote Sensing (http://www2.isprs.org/commissions/comm3/wg4/semantic-labeling.html).

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